Semester IV **B. Tech CSE (AI)**

BCSAI 402: ROBOTIC OPERATING SYSTEMS & ROBOT SIMULATION

3L + 0T + 0P + 3C **MM 100**

Unit-1: Introduction

Understanding ROS, ROS Environment, Installation and Configuration of ROS Environment, ROS Filesystem, ROS package:create and build.

Unit-2: ROS Design: Graph concepts, Nodes, topic, messages, command line tools: roscore, rosnode, rosrun, rostopic, rqt_plot

Unit-3: ROS services: connections and parameters; system dependencies and roslaunch, debugging, visualizing

Unit-4: ROS Standards: coding style, package layout, naming conventions, common procedures

Unit-5: Coordinate frames and transforms: actions/tasks, message ontology, client libraries: main and experimental

Text/Reference Books:

1. Morgan Quigley, Brian Gerkey, William D. Smart, "Programming Robots with ROS", O'Reilly Media,2015
2. Low K. H., "Industrial Robotics: Programming, Simulation and Applications", I-Tech, 2007.
3. John J. Craig, "Introduction to Robotics Mechanics and Control", 3rd edition, Pearson, 2008.
4. B. Siciliano and K.P. Valavanis, "Control problems in Robotics and Automation", Springer, 1998.